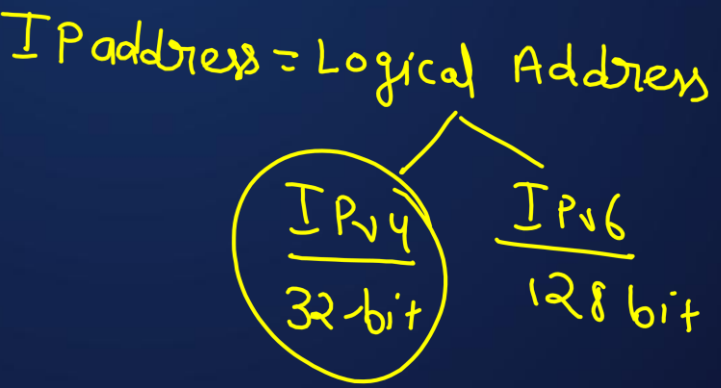
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Data communication & networking Day-4



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Five Different Classes of IPv4 Addresses

Default

Class	First Octet decimal (range)	First Octet binary (range)	IP range	Subnet Mask	Hosts per Network ID	# of networks
Class A	0–127	0XXXXXXX	0.0.0.0-127.255.255.255	255.0.0.0	$2^{24}-2$	2^{7}
Class B	128–191	10XXXXXX	128.0.0.0-191.255.255.255	255.255.0.0	$2^{16}-2$	2^{14}
Class C	192–223	110XXXXX	192.0.0.0-223.255.255.255	255.255.255.0	$2^{8}-2$	2^{21}
Class D (Multicast)	224–239	1110XXXX	224.0.0.0-239.255.255.255			
Class E (Experimental)	240–255	1111XXXX	240.0.0.0-255.255.255.255			

0-255

Loopback

A 0-126(127)

B 128-191

C = 192-223

Public

D = 224-239 = Broadcast

E = 240-255 = Experiment

Ic 192.168.1.200

Net ID, Host ID

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Class	Private address range	
	start address	finish address
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.31.255.255
C	192.168.0.0	192.168.255.255

Class	Public address range	
	start address	finish address
A	0.0.0.0	126.255.255.255
B	128.0.0.0	191.255.255.255
C	192.0.0.0	223.255.255.255
D	224.0.0.0	239.255.255.255
E	240.0.0.0	255.255.255.255

Class A - N · H · H · H

B. N · N · H · H

C = N · N · N · H

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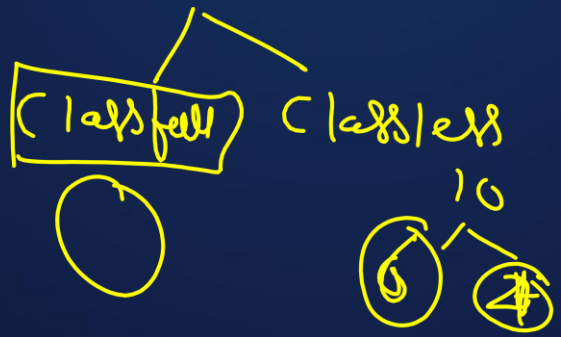
Subnetting in Networking-



In networking,

- The process of dividing a single network into multiple sub networks is called as subnetting.
- The sub networks so created are called as subnets.





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Fixed Length Subnetting-

Fixed length subnetting also called as **classful subnetting** divides the network into subnets where-
All the subnets are of same size.
All the subnets have equal number of hosts.
All the subnets have same subnet mask.

2. Variable Length Subnetting-

Variable length subnetting also called as **classless subnetting** divides the network into subnets where-
All the subnets are not of same size.
All the subnets do not have equal number of hosts.
All the subnets do not have same subnet mask.

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We have a big single network having IP Address 200.1.2.0. class 0-127 0-225 128-255 do subnetting and divide this network into 2 subnets.

1st Part

borrow bit = 1

$2^1 = 2$ Class = C

IP address = 200.1.2.0

N.N.N.H / 8 8 8 8

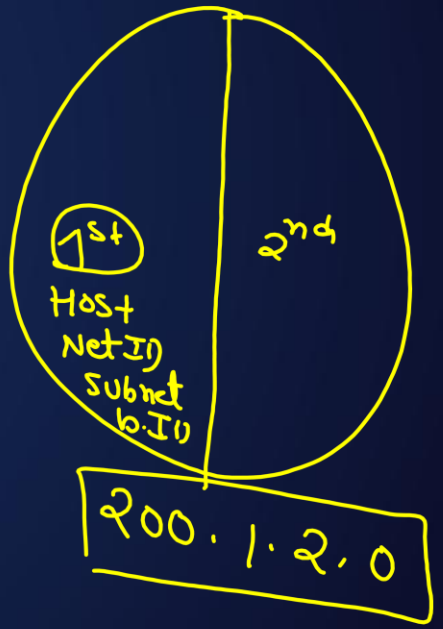
Total No of IP address = $2^7 = 128$ (0-127) 1+7

$2^7 =$

Total no of host config = 128 - 2 = 126 (1-126)

broadcast IP = 200.1.2.127

2nd Subnet broadcast = 200.1.2.255



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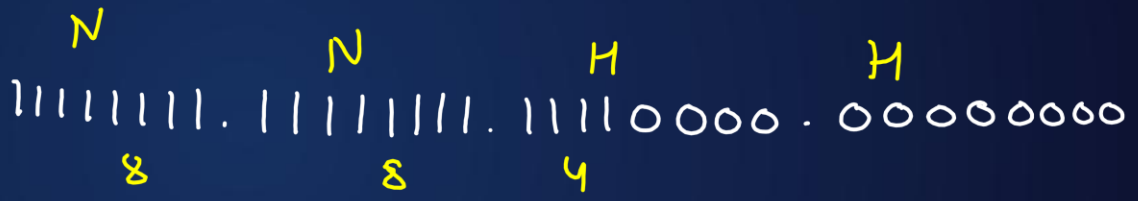
In a class B, network on the internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts per subnet?

4096

4094

4092

4090



32 =

No of bit reserve for host = 32 − 20 = 12

$2^{12} - 2 = 4096 - 2 = 4094$

$2^7 - 2$

No of Subnet possible

borrow bit $2^4 = 16$

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If you have a network with an IP address of 192.168.1.0 and a subnet mask of 255.255.255.128, what are the two possible subnets?

A) 192.168.1.0 and 192.168.1.128

B) 192.168.1.0 and 192.168.1.64

C) 192.168.1.0 and 192.168.0.128

D) 192.168.1.0 and 192.168.1.192

(C)

N N N H

11111111.11111111.11111111.10000000

No of host bit = 2^7 = 128

192.168.1.0 to 192.168.1.127

129.168.1.128 to 192.168.1.255

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What is the broadcast address for the subnet 10.0.0.128/25?

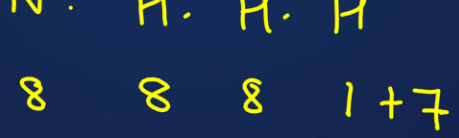
A) 10.0.0.127
B) 10.0.0.255
C) 10.0.0.255
D) 10.0.0.191

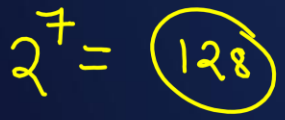














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In a subnet with a mask of 255.255.255.240, how many bits are used for the host portion?

A) 4
B) 12
C) 8
D) 16

11111111.11111111.11111111.11110000

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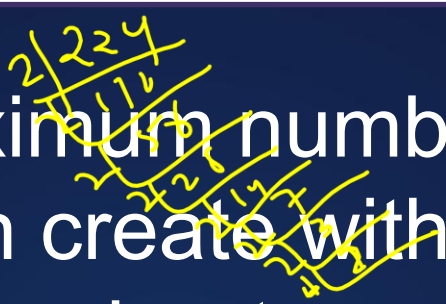
What is the maximum number of subnets you can create with a Class C network using a subnet mask of 255.255.255.224?

A) 4

B) 8

C) 16

D) 32



128 64 32 16 8 4 2 1 0

1 1 1 0 0 0 0 0

No of subnet $= 2^n$ (n = borrow bit)

11111111.11111111.11111111.111 00000

$2^3 = 8$

No of host

total host

Which of the following statements are correct?

(A) IP addresses are used to identify computers on the Internet.

(B) HTTP is used for email sending
(C) Large files such as videos which cannot be sent by e-mail as attachments, can be sent to another computer via FTP.
(D) POP3 is used for file transfer
(E) None of the above

Choose the correct answer from the options given below:

(A) and (B) only
(A) and (C) only
(B) and (C) only
(A), (B), (C)

Which of the following statements are correct?

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Choose the correct answer from the options given below:

(A) and (B) only
(A) and (C) only
(B) and (C) only
(A), (B), (C)

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What operates in the Data Link and the Network layer?

NIC
Bridge
Brouter
Router

Devices on one network can communicate with devices on another network via a

file server
utility server
printer server
gateway

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Which file transfer protocol uses UDP?

NFS
TELNET
TFTP
FTP

Which of the following medium access control technique is used for bus/tree?

token ring
token bus
CSMA
MAC

Which of the following statement is incorrect?

Gateways are assumed to know correct routes; hosts begin with minimal routing information and learn new routes from gateways.
Layered protocols are designed so that layer n at the destination receives exactly the same object sent by layer n at the source.
Application programs as well as all protocol software from the Internet layer upward use only IP addresses; the network interface layer handles physical addresses.
All of the above

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The physical layer of a network

defines the electrical characteristics of signals passed between the computer and communication devices
controls error detection and correction
constructs packets of data and sends them across the network
All of the above

An error detecting code is which code is the remainder resulting from dividing the bits to be checked by a predetermined binary number, is known as

Cyclic redundancy check
Checksum
Error detecting code
Error rate

In OSI model, which of the following layer provides error-free delivery of data?

Data link
Network
transport
Session

Error detection at the data link level is achieved by?

Bit stuffing
Hamming codes
Cyclic Redundancy codes
Equalization

What is the minimum number of wires needed to send data over a serial communication link layer?

1
2
4
6

The _____ layer is the layer closest to the transmission medium.

transport
network
data link
physical

In a _____ topology, if there are n devices in a network, each device has n - 1 ports for cables.

ring
bus
star
mesh

The network layer, in reference to the OSI model, provide

data link procedures that provide for the exchange of data via frames that can be sent and received
the interface between the X.25 network and packet mode device
the virtual circuit interface to packet-switched service
All of the above

Which of the following communications lines is best suited to interactive processing applications?

Half duplex
simplex lines
full-duplex lines
mixed band channels

The 32-bit internet address 10000000 00001010 00000010 00011110 will be written in dotted decimal notation as

148.20.2.30
164.100.9.61
210.20.2.64
128.10.2.30

Which of the following is the way in which an IOT device is associated with data.

Internet
Cloud
Automata
Network

2. This is a standard for security that is located at the network or packet-processing layer of or packet-processing layer of network communications, as opposed to the application layer.

a) SSL

b) TCP

c) S-HTTP

d) passive FTP

e) IPsec

The default subnet mask for a class B network can be

(A) 255.255.255.0
(B) 255.0.0.0
(C) 255.255.192.0
(D) 255.255.0.0

The default subnet mask for a class B network can be

(A) 255.255.255.0
(B) 255.0.0.0
(C) 255.255.192.0
(D) 255.255.0.0

Firewall is a device that filters access to the protected network from the outside network. Firewall can filter the packets on the basis of

(A) Source IP address
(B) Destination IP Address
(C) TCP Source Port
(D) UDP Source Port
(E) TCP Destination Port

Choose the correct answer from the options given below:
i ➥ (A), (B) and (C) only
ii ➥ (B) and (E) only
iii ➥(C) and (D) only
iv ➥(A), (B), (C), (D) and (E) only

End-to-end connectivity is provided from host-to-host in:

Network layer
Session layer
Data link layer
Transport layer
None of the above

CSMA (Carrier Sense Multiple Access) is

a method of determining which device has access to the transmission medium at any time
a method access control technique for multiple-access transmission media.
a very common bit-oriented data link protocol issued by ISO.
network access standard for connecting stations to a circuit-switched network

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- **What key(s) are used by the sender of an encrypted message in an asymmetric-key cipher?**

- Neither public nor private key of the receiver
- Public key of the receiver
- Public or private key of the sender
- Private key of the sender
- None of the above

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- **Symmetric Key Cryptography uses stream cipher to encrypt the information. One of example of stream cipher:**

- SHA
- RC4
- MD5
- Blowfish
- None of the above

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- In computing, _____ is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

- Spyware
- Cookie
- Spam
- Firewall

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- What is the term for a cyber-security attack that targets multiple interconnected devices simultaneously to create a large- scale attack network?

- DDoS attack
- Botnet attack
- Zero-day attack
- Spear phishing attack

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- **Routing involves:**

- Dividing data into frames
- Managing flow control
- Determining the best path for data packets
- Correcting errors in data transmission
- None of the above

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- Which of the following statement/s is/are NOT CORRECT?
- (A) OSPF is based on distance-vector routing protocol.
- (B) Both link-state and distance-vector routing are based on the least cost goal.
- (C) BGP4 is based on the path-vector algorithm.
- (D) The three-node instability can be avoided using split horizon combined with poison reverse.
- (E) RIP is based on link state algorithm.
- Choose the correct answer from the options given below:
- (A), (D) and (E) Only

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- **What does Router do in a network?**

- Forwards a packet to all outgoing links
- Forwards a packet to the next free outgoing link
- Determines on which outing link a packet is to be forwarded
- Forwards a packet to all outgoing links except the originated link
- None of the above

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- _____ serves as the entry and exit point of a network, as all data coming in or going out of a network must first pass through it in order to use routing paths.

- Bridge
- Router
- Gateway
- Repeater
- None of the above

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- **Which one of the following fields of an IP header is NOT modified by a typical IP router?**

- Checksum
- Source address
- Time to Live (TTL)
- Length
- None of the above

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- **What is routing algorithm used by OSPF routing protocol?**

- Distance vector
- Flooding
- Path vector
- Link state
- None of the above

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- Host A(on TCP/IP v4 network A) sends an IP datagram D to host B (also on TCP/IP v4 network B) Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D?

- i) TTL ii) Checksum iii) Fragment Offset

- (i) only
- (i) and (ii) only
- (ii) and (iii) only
- (i), (ii) and (iii)

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- Consider the following statements about the functionality of an IP based router.

- I. A router does not modify the IP packets during forwarding.

- II. It is not necessary for a router to implement any routing protocol.

- III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.

- Which of the above statements is/are TRUE?
- I and II only
- I only
- II and III only
- II only

Given Ethernet address 01011010 00010001 01010101 00011000 10101010 00001111 in binary, what is the address in hexadecimal notation ?

i ➥ 5A : 88 : AA : 18 : 55 : F0

ii ➥ 5A : 81 : BA : 81 : AA : 08

iii ➥ 5A : 18 : 5A : 18 : 55 : 0F

iv ➥ 5A : 11 : 55 : 18 : 55 : 0F

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List (I)		List(II)	
A.	CDMA	I.	It Provides mobile internet connection with faster data transfer rate
B.	GSM	II.	It allow uses to connect to a network or to other devices over wireless channel
C.	UMTS	III.	Accessing mechanism for multiple transmitters over a single channel
D.	Wi-Fi	IV.	It is cellular technology employs hybrid of FDMA and TDMA

Choose the correct answer from the options given below:

i ➥ A-(III), B-(IV), C-(II), D-(I)
ii ➥ A-(III), B-(IV), C-(I), D-(II)
iii ➥ A-(II), B-(III), C-(IV), D-(I)
iv ➥ A-(II), B-(I), C-(IV), D-(III)

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